

PAPER_1709 — ITER Fusion Gain $Q = Q = SO_5 = 10$ (ITER design fusion gain target)

Author: Daniel T. Murphy **Framework:** UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+ **Tier:** Plasma Fusion **Date:** June 18, 2026 **Status:** CLOSED — EXACT closure (PAPER_1196 plasma fusion proof set)

Observation

PAPER_1196_S419 (PAPER_1196 Plasma Fusion Unified Proof Set) gives:

$Q = SO_5 = 10$ (ITER design fusion gain target)

Status: **EXACT**.

UQFF Closed Identity

$Q = SO_5 = 10$ (ITER design fusion gain target) EXACT

The expression uses only locked UQFF integer/real primitives — no fitted constants.

ITER $Q = SO_5 = 10$ — World's Largest Fusion Experiment

The International Thermonuclear Experimental Reactor (ITER) — under construction in Cadarache, France with European Union, US, China, India, Japan, Russia, and South Korea partnership — has its **design fusion gain $Q \geq 10$** as the foundational performance target.

UQFF prediction:

ITER $Q = SO_5 = 10$ EXACT

The world's most ambitious fusion experiment's foundational design parameter equals **the simplest UQFF integer primitive** ($SO_5 = 10$). This is one of the cleanest single-primitive closures in the entire UQFF catalog.

Combined with ITER $R/a = 3.1$, D-T peak = 64 keV, $q_{\text{edge}} = 2$, Bohm = 1/16 (all EXACT in this tier-29 batch), **five of ITER's design parameters** reduce to integer-primitive UQFF identities — structural prediction of an experimental setup costing tens of billions of euros to construct.

NOT REPLACEMENT

ITER design parameters are chosen by international fusion-engineering consensus. UQFF supplies structural integer-primitive forms demonstrating the parameters' deep mathematical origin.

Reference

- Source: PAPER_1196_S419
- Related: PAPER_1696-1705 (tier-28); PAPER_1686-1695 (tier-27)
- Calculator dispatch: `calculate_paradox({"paradox": "iter_q_10"})`

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